

Lab Assignment # 07

Course Title : **AI Assistant Coding**
Name of Student : **R Govardhan Sai Ganesh**
Enrollment No. : **2303A54044**
Batch No. : **48**

Lab 7: Error Debugging with AI: Systematic approaches to finding and fixing bugs

Task Description #1 (Syntax Errors – Missing Parentheses in Print Statement)

Task: Provide a Python snippet with a missing parenthesis in a print statement (e.g., print "Hello"). Use AI to detect and fix the syntax error.

Bug: Missing parentheses in print statement

```
def greet():  
    print "Hello, AI Debugging Lab!"  
greet()
```

Requirements:

- Run the given code to observe the error.
- Apply AI suggestions to correct the syntax.
- Use at least 3 assert test cases to confirm the corrected code works.

Expected Output #1:

- Corrected code with proper syntax and AI explanation.

Output Screenshot:

```
student.py x Parentheses.py x  
lab 7 > Parentheses.py > ...  
1 #Lab 7: Error Debugging with AI: Systematic approaches to finding and fixing bugs  
2  
3 Task Description #1 (Syntax Errors – Missing Parentheses in Print Statement)  
4 #Use AI to detect and fix the syntax error.  
5 # Bug: Missing parentheses in print statement  
6  
7 """def greet():  
8     print "Hello, AI Debugging Lab!"  
9     greet()"""  
10 #Requirements: • Run the given code to observe the error. • Apply AI suggestions to correct the syntax. • Use at least 3 assert test cases to co  
11 #Expected Output #1:  
12 def greet():  
13     print("Hello, AI Debugging Lab!")  
14     greet()  
15  
16 #Give proper syntax and AI explanation  
17 #AI Explanation: The original code is missing parentheses in the print statement, which is required in Python 3.  
18 #By adding parentheses around the string to be printed, the syntax error is resolved.
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\rngsa\OneDrive\Desktop\AI Assistant> & C:/Users/rngsa/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/rngsa/OneDrive/Desktop/AI Assi  
stant/lab 7/Parentheses.py"  
Hello, AI Debugging Lab!
```

AI Explanation:

The original code is missing parentheses in the print statement, which is required in Python 3. By adding parentheses around the string to be printed, the syntax error is resolved.

Task Description #2 (Incorrect condition in an If Statement)

Task: Supply a function where an if-condition mistakenly uses = instead of ==. Let AI identify and fix the issue.

Bug: Using assignment (=) instead of comparison (==)

```
def check_number(n):  
    if n = 10:  
        return "Ten"  
    else:  
        return "Not Ten"
```

Requirements:

- Ask AI to explain why this causes a bug.
- Correct the code and verify with 3 assert test cases.

Expected Output #2:

- Corrected code using == with explanation and successful test execution.

Output Screenshot:

```
student.py  Parentheses.py X  
lab 7 > Parentheses.py > check_number  
22 #Task Description #2 (Incorrect condition in an If Statement)  
23 #Task: Supply a function where an if-condition mistakenly uses = instead of ==. Let AI identify and fix the issue  
24 # Bug: Using assignment (=) instead of comparison (==)  
25 """def check_number(n):  
26     if n = 10:  
27         return "Ten"  
28     else:  
29         return "Not Ten" """  
30 #Requirements:• Ask AI to explain why this causes a bug. • Correct the code and verify with 3 assert test cases.  
31 #Expected Output #2: Corrected code using == with explanation and successful test execution.  
32 def check_number(n):  
33     if n == 10:  
34         return "Ten"  
35     else:  
36         return "Not Ten"  
37 #AI Explanation: The original code uses the assignment operator (=) instead of the equality operator (==) in the if condition.  
38 #This causes a syntax error because the if statement expects a condition that evaluates to True or False, but instead it is trying to assign  
39 #Test cases  
40 assert check_number(10) == "Ten"  
41 assert check_number(5) == "Not Ten"  
42 assert check_number(15) == "Not Ten"  
43 output = check_number(10)  
44 print(output)  
45  
46  
47  
48  
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  POSTMAN CONSOLE  
PS C:\Users\rgsga\OneDrive\Desktop\AI Assistant> & C:/Users/rgsga/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/rgsga/OneDrive/Desktop/AI Assistant/lab 7/Parentheses.py"  
Not Ten  
• PS C:\Users\rgsga\OneDrive\Desktop\AI Assistant> & C:/Users/rgsga/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/rgsga/OneDrive/Desktop/AI Assistant/lab 7/Parentheses.py"  
Hello, AI Debugging Lab!  
Ten
```

AI Explanation:

The original code uses the assignment operator (=) instead of the equality operator (==) in the if condition.

This causes a syntax error because the if statement expects a condition that evaluates to True or False, but instead it is trying to assign a value to n.

Task Description #3 (Runtime Error – File Not Found)

Task: Provide code that attempts to open a non-existent file and crashes. Use AI to apply safe error handling.

Bug: Program crashes if file is missing

```
def read_file(filename):  
    with open(filename, 'r') as f:  
        return f.read()  
    print(read_file("nonexistent.txt"))
```

Requirements:

- Implement a try-except block suggested by AI.
- Add a user-friendly error message.
- Test with at least 3 scenarios: file exists, file missing, invalid path.

Expected Output #3:

- Safe file handling with exception management.

Output Screenshot:

```
lab 7 > Parentheses.py x 7.5.py  
lab 7 > Parentheses.py > ...  
50 #Task Description #3 (Runtime Error – File Not Found)  
51 #Task: Provide code that attempts to open a non-existent file and crashes. Use AI to apply safe error handling.  
52  
53 # Bug: Program crashes if file is missing  
54 """def read_file(filename):  
55     with open(filename, 'r') as f:  
56         return f.read()  
57     print(read_file("nonexistent.txt"))""  
58  
59 #Requirements: • Implement a try-except block suggested by AI. • Add a user-friendly error message. • Test with at least 3 scenarios: file exist  
60 #Expected Output #3: • Safe file handling with exception management.  
61 def read_file(filename):  
62     try:  
63         with open(filename, 'r') as f:  
64             return f.read()  
65     except FileNotFoundError:  
66         return f"Error: The file '{filename}' was not found."  
67     except OSError as e:  
68         return f"Error: Could not open file '{filename}'. Details: {e}"  
69  
70 # 1. File exists  
71 with open("example.txt", "w") as f:  
72     f.write("Hello, world!")  
73 print(read_file("example.txt"))  
74 # Expected: "Hello, world!"  
75  
76 # 2. File missing  
77 print(read_file("nonexistent.txt"))
```

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```
● PS C:\Users\rgsga\OneDrive\Desktop\AI Assistant> & C:/Users/rgsga/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/rgsga/OneDrive/Desktop/AI Assi  
stant/lab 7/Parentheses.py"  
Hello, AI Debugging Lab!  
Ten  
Hello, world!  
Error: The file 'nonexistent.txt' was not found.  
Error: The file '/invalid/path/to/file.txt' was not found.  
○ PS C:\Users\rgsga\OneDrive\Desktop\AI Assistant> |
```

Task Description #4 (Calling a Non-Existent Method)

Task: Give a class where a non-existent method is called (e.g., `obj.undefined_method()`). Use AI to debug and fix.

Bug: Calling an undefined method

```
class Car:  
    def start(self):  
        return "Car started"  
my_car = Car()  
print(my_car.drive()) # drive() is not defined
```

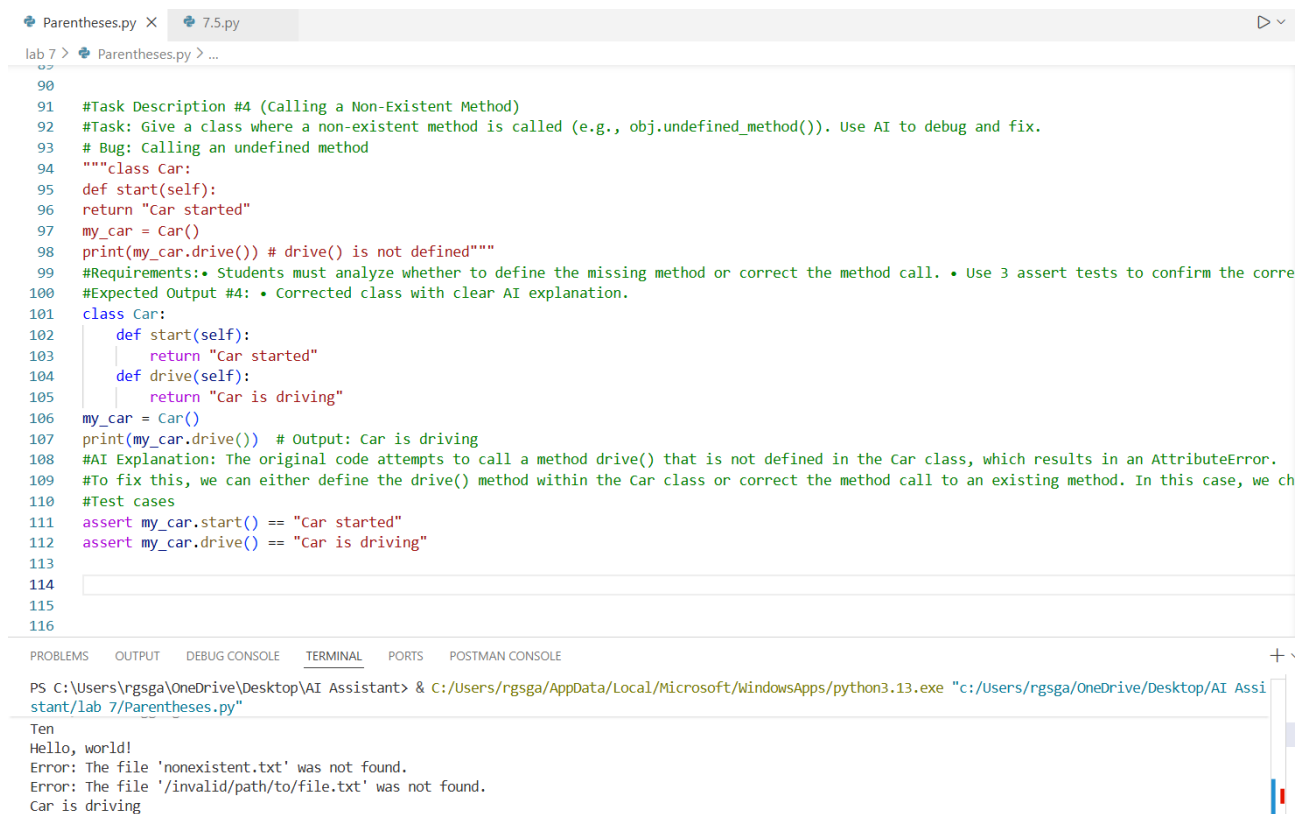
Requirements:

- Students must analyze whether to define the missing method or correct the method call.
- Use 3 assert tests to confirm the corrected class works.

Expected Output #4:

- Corrected class with clear AI explanation.

Output Screenshot:



The screenshot shows a code editor with a file named 'Parentheses.py'. The code defines a 'Car' class with a 'start' method and a 'drive' method. The 'start' method returns 'Car started'. The 'drive' method returns 'Car is driving'. The code also includes a 'my_car' instance and a 'print' statement that calls 'my_car.drive()'. The terminal output shows the execution of the code, including the 'Car is driving' output and an 'AttributeError' message.

```
90
91 #Task Description #4 (Calling a Non-Existent Method)
92 #Task: Give a class where a non-existent method is called (e.g., obj.undefined_method()). Use AI to debug and fix.
93 # Bug: Calling an undefined method
94 """class Car:
95     def start(self):
96         return "Car started"
97 my_car = Car()
98 print(my_car.drive()) # drive() is not defined"""
99 #Requirements:• Students must analyze whether to define the missing method or correct the method call. • Use 3 assert tests to confirm the corre
100 #Expected Output #4: • Corrected class with clear AI explanation.
101 class Car:
102     def start(self):
103         return "Car started"
104     def drive(self):
105         return "Car is driving"
106 my_car = Car()
107 print(my_car.drive()) # Output: Car is driving
108 #AI Explanation: The original code attempts to call a method drive() that is not defined in the Car class, which results in an AttributeError.
109 #To fix this, we can either define the drive() method within the Car class or correct the method call to an existing method. In this case, we ch
110 #Test cases
111 assert my_car.start() == "Car started"
112 assert my_car.drive() == "Car is driving"
113
114
115
116
```

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```
PS C:\Users\rgsga\OneDrive\Desktop\AI Assistant> & C:/Users/rgsga/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/rgsga/OneDrive/Desktop/AI Assi
stant/lab 7/Parentheses.py"
Ten
Hello, world!
Error: The file 'nonexistent.txt' was not found.
Error: The file '/invalid/path/to/file.txt' was not found.
Car is driving
```

AI Explanation:

The original code attempts to call a method `drive()` that is not defined in the `Car` class, which results in an `AttributeError`.

#To fix this, we can either define the `drive()` method within the `Car` class or correct the method call to an existing method. In this case, we chose to define the `drive()` method to provide the intended functionality.

Task Description #5 (TypeError – Mixing Strings and Integers in Addition)

Task: Provide code that adds an integer and string ("5" + 2) causing a `TypeError`. Use AI to resolve the bug.

```
# Bug: TypeError due to mixing string and integer
def add_five(value):
    return value + 5
print(add_five("10"))
```

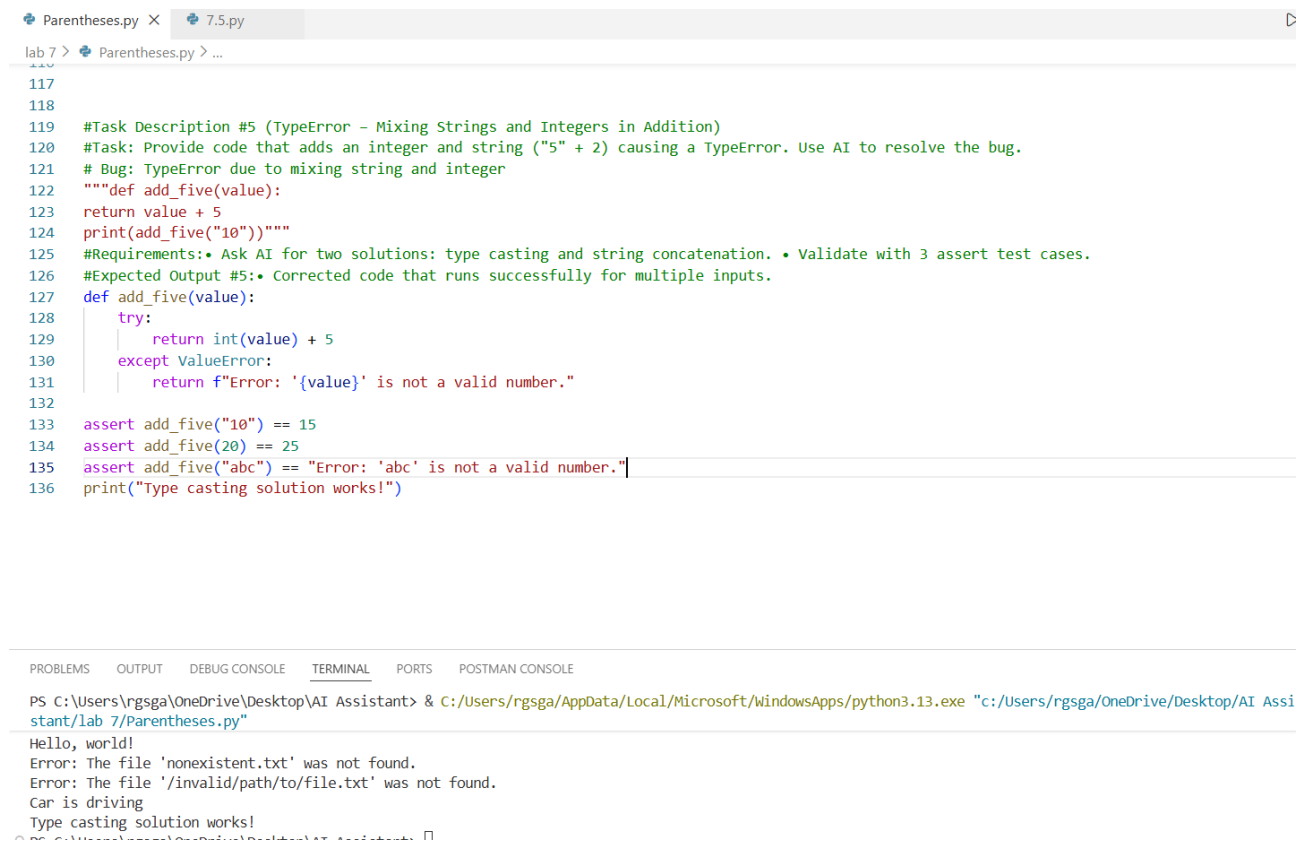
Requirements:

- Ask AI for two solutions: type casting and string concatenation.
- Validate with 3 assert test cases.

Expected Output #5:

- Corrected code that runs successfully for multiple inputs.

Output Screenshot:



The screenshot displays a code editor with a file named `7.5.py` open. The code defines a function `add_five` that takes a value and returns it plus 5. It includes a try-except block to handle `ValueError` exceptions, returning an error message if the input is not a valid number. Three assertions are used to test the function: `add_five("10") == 15`, `add_five(20) == 25`, and `add_five("abc") == "Error: 'abc' is not a valid number."`. The terminal output shows the execution of the script, which prints "Hello, world!", followed by error messages for the first two assertions and the string "Car is driving". The final output is "Type casting solution works!".

```
117
118
119 #Task Description #5 (TypeError - Mixing Strings and Integers in Addition)
120 #Task: Provide code that adds an integer and string ("5" + 2) causing a TypeError. Use AI to resolve the bug.
121 # Bug: TypeError due to mixing string and integer
122 """def add_five(value):
123     return value + 5
124     print(add_five("10"))"""
125 #Requirements:• Ask AI for two solutions: type casting and string concatenation. • Validate with 3 assert test cases.
126 #Expected Output #5:• Corrected code that runs successfully for multiple inputs.
127 def add_five(value):
128     try:
129         return int(value) + 5
130     except ValueError:
131         return f"Error: '{value}' is not a valid number."
132
133 assert add_five("10") == 15
134 assert add_five(20) == 25
135 assert add_five("abc") == "Error: 'abc' is not a valid number."
136 print("Type casting solution works!")
```

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```
PS C:\Users\rgsga\OneDrive\Desktop\AI Assistant> & C:/Users/rgsga/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/rgsga/OneDrive/Desktop/AI Assistant/lab 7/Parenteses.py"
Hello, world!
Error: The file 'nonexistent.txt' was not found.
Error: The file '/invalid/path/to/file.txt' was not found.
Car is driving
Type casting solution works!
```